Alessandro **Petruzzelli**

AI Researcher · PhD Candidate in Computer Science - Artificial Intelligence

Department of Computer Science - University of Bari Aldo Moro

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Research & Professional Experience

University of Bari — *Bari, Italy*

PhD Researcher — *Oct 2023 - Dec 2026*

- Developing frameworks to inject domain-specific knowledge into LLMs, demonstrating significant improvements in recommendation accuracy.
- Designing and building end-to-end conversational AI systems capable of multi-turn, personalized interactions.
- Pioneering research on symbiotic systems, aiming to create more transparent and user-centric AI.

University of Bari — *Bari, Italy*

Research Fellow — *Feb 2023 - Oct 2023*

- Conducted a formal analysis and validation of recommendation algorithms for retail personalization.
- Authored a comprehensive report on state-of-the-art methods, influencing subsequent project directions.

Education

University of Bari — *Bari, Italy*

Ph.D. in Computer Science - Artificial Intelligence — *Oct 2023 - Dec 2026*

Thesis: Knowledge Injection and User-Behavior Modeling in Large Language Model-Based Recommender Systems

University of Bari — *Bari, Italy*

M.S. in Data Science — *Oct 2021 - Jul 2023*

Thesis: Transformer-Based Conversational Recommendation based on Knowledge Graphs

University of Bari — *Bari, Italy*

B.S. in Computer Science and Software Production Technology — *Oct 2018 - Jul 2021*

Thesis: Design and Development of a Content-based Recommender System to Support a Conversational Agent

Skills

GenAI & LLM Ecosystem	LangChain, Hugging Face Transformers, RAG, Prompt Engineering, PyTorch, TensorFlow
Programming & Data	Python (Expert), Java (Proficient), R (Familiar), SQL (Familiar)
Distributed AI & MLOps	DeepSpeed, PyTorch Fully Sharded Data Parallel (FSDP), Git, Docker
Languages	Italian (Native), English (Professional Working Proficiency)

Selected Publications

Journal Articles

Integrating Heterogeneous Knowledge for Enhanced Recommendation with Large Language Models — *ACM Transactions on Information Systems (TOIS)*

Petruzzelli, A., Musto, C., de Gemmis, M., Lops, P., & Semeraro, G. — *2026*

- Proposes REKALM, a framework that lexicalizes heterogeneous knowledge sources into natural language and aligns LLM-based recommenders with domain-specific information through knowledge-aware instruction tuning.

Beyond the Lookup: Simulating Realistic User Uncertainty for the Evaluation of Conversational Agentic Recommenders — *Information Systems Frontiers*

Petruzzelli, A., Martina, A. F. M., Musto, C., de Gemmis, M., et al. — *2026*

- Introduces open-weight user simulators, trained via teacher–student distillation, that reproduce ambiguous user behavior and reveal a robustness gap in state-of-the-art agentic conversational recommenders.

Conference Papers

Simulating Diverse User Behavioral Stereotypes for Evaluating Agentic Conversational Recommenders — *The 20th ACM Conference on Recommender Systems (RecSys)*

Petruzzelli, A., Martina, A. F. M., Musto, C., de Gemmis, M., et al. — *Minneapolis, Minnesota, USA, 2026*

- Introduces a stereotype-based user simulator built via knowledge distillation, showing that diverse behavioral stereotypes expose the limitations of current evaluation practices for agentic recommender systems.

DistillRecDial: A Knowledge-Distilled Dataset Capturing User Diversity in Conversational Recommendation

The 19th ACM Conference on Recommender Systems (RecSys)

Martina, A. F. M., **Petruzzelli, A.**, Musto, C., de Gemmis, M., et al.

Prague, Czech Republic, 2025

- Introduces a novel dataset generated via knowledge distillation to simulate diverse user stereotypes, improving dialogue quality and realism for training CRSs.

Empowering Recommender Systems based on Large Language Models through Knowledge Injection Techniques

The 33rd ACM Conference on User Modeling, Adaptation and Personalization (UMAP)

Petruzzelli, A., Musto, C., de Gemmis, M., Semeraro, G., & Lops, P.

New York City, NY, USA, 2025

- Proposes a novel framework for injecting domain-specific knowledge into LLMs to enhance recommendation accuracy and relevance.

Instructing and Prompting Large Language Models for Explainable Cross-domain Recommendations

The 18th ACM Conference on Recommender Systems (RecSys)

Petruzzelli, A., Musto, C., Laraspata, L., et al.

Bari, Italy, 2024

- Investigates prompt engineering and fine-tuning strategies that enable LLMs to provide accurate cross-domain recommendations with natural language explanations.

Improving Transformer-based Sequential Conversational Recommendations through Knowledge Graph Embeddings

The 32nd ACM Conference on User Modeling, Adaptation and Personalization (UMAP)

Petruzzelli, A., Martina, A. F. M., Spillo, G., et al.

Cagliari, Sardinia, Italy, 2024

- Demonstrates that integrating knowledge graph embeddings into Transformer models significantly boosts performance in sequential conversational recommendation scenarios.

Preprints

Generative User Simulators in Recommendation: A Survey

SSRN preprint

Petruzzelli, A., Musto, C., & Jannach, D.

2026

- Surveys generative user simulators for recommendation, proposing a conceptual architecture to categorize existing approaches and reviewing how their behavioral fidelity is evaluated.

Academic Service & Committees

20th ACM Conference on Recommender Systems (RecSys '26)

Minneapolis, Minnesota, USA

Program Committee Member

2026

- Full & Short Papers
- Late Breaking Results
- Reproducibility Track

The ACM Web Conference 2026 (WWW '26)

Dubai, United Arab Emirates

Program Committee Member

2026

19th ACM Conference on Recommender Systems (RecSys '25)

Prague, Czech Republic

Program Committee Member

2025

- Full & Short Papers
- Late Breaking Results
- Reproducibility Track
- Workshop: Recommender Systems for Sustainability and Social Good (RecSoGood)
- Workshop: Interfaces and Human Decision Making for Recommender Systems (IntRS)

33rd ACM Conference on User Modeling, Adaptation and Personalization (UMAP '25)

New York City, NY, USA

Program Committee Member

2025

- Full & Short Papers
- Late Breaking Results & Demo Papers
- Workshop: Explainable User Models and Personalized Systems (ExUM)

18th ACM Conference on Recommender Systems (RecSys '24)

Bari, Italy

Program Committee Member

2024

- Full & Short Papers
- Workshop: Recommender Systems for Sustainability and Social Good (RecSoGood)

32nd ACM Conference on User Modeling, Adaptation and Personalization (UMAP '24)

Cagliari, Sardinia, Italy

Program Committee Member

2024

- Full & Short Papers
- Workshop: Explainable User Models and Personalized Systems (ExUM)

Extracurricular Activities

Advanced Training & Schools

ACM RecSys, SIGCHI, ACM Europe, EURAI, and University of Cagliari — *Bari, Italy*

ACM Europe School on Recommender Systems — *Oct 2024*

Association for Logic, Language and Information — *Leuven, Belgium*

35th European Summer School in Logic, Language and Information (ESSLLI) — *Aug 2024*

Associazione Italiana di Linguistica Computazionale — *Bari, Italy*

Lectures on Computational Linguistics — *Jul 2024*

Various Sponsors — *Bari, Italy*

9th International School on Deep Learning — *Apr 2023*

Teaching

Short Master in Generative AI — *Bari, Italy*

Lecturer — *Jan 2026 - Feb 2026*

- Theoretical and hands-on lessons on improving LLM performance through Retrieval-Augmented Generation (RAG).

SPEGEA Business School — *Bari, Italy*

Lecturer — *Jul 2024 - Sep 2024*

- Designed and delivered a 6-hour professional course titled "How to Build a Chatbot".

Selected Presentations

Oral Presentation at the 33rd ACM Conference on User Modeling, Adaptation and Personalization (UMAP '25) — *New York City, NY, USA*

Empowering Recommender Systems based on LLMs through Knowledge Injection Techniques — *Jun 2025*

- Presented a framework for improving LLM-based recommenders by injecting structured domain knowledge, showing significant gains in performance.

Oral Presentation at the 18th ACM Conference on Recommender Systems (RecSys '24) — *Bari, Italy*

Instructing and Prompting LLMs for Explainable Cross-domain Recommendations — *Oct 2024*

- Detailed our investigation into prompt engineering strategies for generating accurate and explainable cross-domain recommendations from LLMs.

Presentation at the RecSys '24 Doctoral Symposium — *Bari, Italy*

Towards Symbiotic Recommendations: Leveraging LLMs for Conversational Recommendation Systems — *Oct 2024*

- Presented my doctoral research vision on creating symbiotic conversational systems to an audience of senior researchers and peers.